Open-VisualJev: Design and Analysis of a Multimodal Decision Model

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Abstract

Multimodal decision models aim to support low-latency decisions by directly mapping visual context and runtime-defined choices to probabilities. Building such models requires a scoring interface that accommodates changing candidate sets and produces reliable decision scores. We present Open-VisualJev, a multimodal decision model built on a vision–language backbone with a dynamic candidate-scoring head. The model combines independent candidate encoding with permutation-equivariant set interaction, enabling joint comparison of the available choices while preserving consistency under candidate reordering. Calibration-aware training and held-out temperature scaling refine the resulting probabilities. Experiments on visual relation tasks, fine-grained recognition, and natural visual question answering show that this design preserves decisions under candidate reordering, learns set-dependent preferences, and remains competitive with a complete-candidate language-model readout on standard tasks. Its advantage becomes more pronounced with larger difficult candidate sets. Component analyses further clarify the contributions of candidate interaction, variable-count training, and probability calibration. These findings support explicit candidate-set modeling as a practical design principle for multimodal decision systems.

1 Introduction

Vision–language models provide a general-purpose foundation for interpreting visual observations through natural-language instructions. Their integration into decision pipelines introduces a distinct inference requirement: converting this semantic understanding into low-latency, machine-consumable judgments. In this setting, the application specifies a decision space at runtime, and the model assigns probabilities to its alternatives. The emerging System One paradigm motivates direct prediction for such judgments, with visual decision models extending this interface to image-conditioned tasks (Yu & Yao, 2026; Zhou et al., 2026). This formulation brings the semantic flexibility of pretrained vision–language models to a structured probabilistic interface.

A central architectural question is how the runtime candidate set should enter the scoring function. Its elements have semantic content, but their presentation order is usually incidental to the decision. At the same time, the appropriate comparison can depend on which alternatives are available. A scorer should therefore preserve candidate identity under reordering while allowing relative preferences to change with the set. These requirements distinguish presentation symmetry from semantic interaction. Independent candidate scores provide the former, while joint comparison supplies the capacity needed for set-dependent decisions. Set representation learning offers a principled foundation for combining these properties (Zaheer et al., 2017; Lee et al., 2019).

We present Open-VisualJev[1], a multimodal decision model built on a vision–language backbone with a dynamic candidate-scoring head. The architecture separates visual grounding of individual candidates from comparison within the current set. Shared-parameter

[1] An independently developed research implementation of a Jev-style multimodal decision model.

branches encode each candidate conditioned on the image and question; a position-free set module then models candidate interactions, and a shared pointer maps the resulting representations to a probability distribution over the supplied choices. This composition makes the scoring function permutation equivariant while retaining set-dependent preferences. Calibration-aware training and held-out temperature scaling complete the probability estimation pipeline. Figure 1 presents the model.

This design provides a basis for analyzing decision models through their observable behavior. Candidate reordering tests whether scores follow candidate identity; preference-reversal tasks test whether the model can revise relative preferences as the set changes. We connect these structural tests to standard task performance using a common Qwen3-VL backbone. A unified six-dataset recognition study evaluates accuracy and robustness to candidate-count changes, while separately supervised natural VQA tests the scoring architecture beyond recognition. Component comparisons and calibration ablations examine how interaction and probability estimation contribute to the complete model.

The experiments show that Open-VisualJev preserves its decisions throughout the exhaustive permutation evaluation and learns the set-dependent preferences required by the relation tasks. It remains competitive with the complete-candidate LM on standard recognition, with a larger advantage on expanded sets of difficult candidates, and shows little accuracy degradation relative to the native LM on natural VQA. Calibration analyses further support combining training-time confidence regularization with post-training temperature scaling. Together, these results connect the candidate-scoring architecture to decision stability, comparative capacity, and probability quality.

Our contributions are threefold:

- We develop a multimodal decision model that combines pretrained visual–language representations with dynamic candidate scoring and an explicit probability-calibration pipeline.
- We formulate a candidate-scoring architecture that combines permutation-equivariant set interaction with shared pointer readout, and analyze its order stability and capacity for set-dependent preferences.
- We evaluate these properties through controlled structural tests, unified fine-grained recognition, and natural VQA, with ablations that characterize candidate interaction, candidate-count training, and probability calibration.

2 Decision interface and model structure

For an image x, a question q, and runtime-supplied candidate texts $C = (c_i)_{i=1}^{K}$, the model returns a normalized distribution over the current candidate slots. External question and candidate identifiers associate outputs with inputs; pointer prompts contain the image, question, and candidate text. This work studies mutually exclusive visual Choice decisions. The output distribution is defined over the supplied candidate set, and a held-out calibration stage adjusts its confidence.

2.1 Candidate encoding and set-conditioned scoring

The scorer first grounds each candidate in the image and question, then compares the resulting representations as a set. This separates the visual interpretation of an alternative from its role among the available choices.

Shared candidate branches. Let E_θ denote the image–text backbone and its final valid-token representation. We construct a question representation $u = E_\theta(x, q)$ and a candidate representation $h_i = E_\theta(x, q \parallel \texttt{Candidate:} \parallel c_i)$ using shared parameters. Each branch is encoded independently of the other candidate texts. The language backbone uses causal attention within each branch. The implementation batches one question branch and one branch per candidate, each with its own token history, multimodal positions, attention mask,

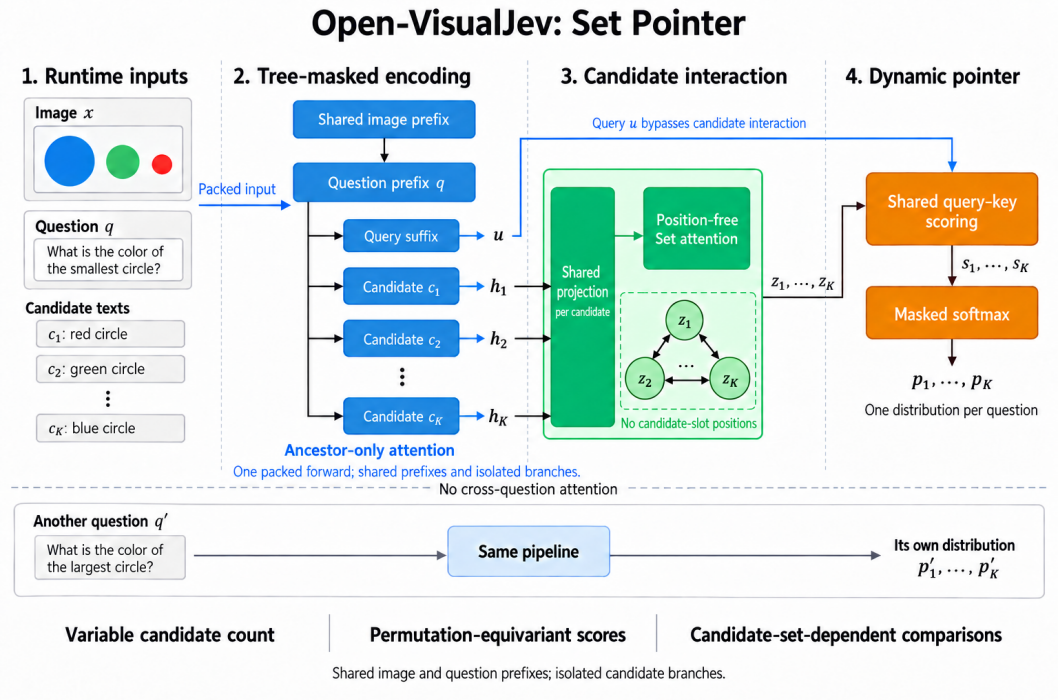


Figure 1: Set Pointer with tree-masked encoding. A packed input shares the image prefix and each question prefix across isolated query and candidate suffixes. Each token attends to its causal predecessors within its branch and to its ancestors. The shared projection is applied independently to each candidate representation; position-free Set attention then compares candidates jointly. The question query bypasses candidate interaction and scores the resulting keys before masked softmax. Each question has its own interaction and normalization. This diagram depicts the tree execution option; the main experiments use ordinary batched branches with the same visibility structure. Training includes dataset-wise mean-confidence regularization, followed by held-out temperature scaling.

and final valid-token readout. Parameters are shared across branches; image encoding is performed within each branch.

Position-free candidate interaction. Writing the candidate representations as rows of H, we apply a shared input projection followed by a position-free candidate module S_ϕ. The Set module uses self-attention over candidate representations, followed by shared per-row transformations. The query representation remains separate from this candidate self-attention stack. Candidate logits and probabilities are

$$Z = S_\phi(H), \qquad s_i = \frac{\langle W_q \,\mathrm{LN}(u), W_k \,\mathrm{LN}(Z_i)\rangle}{\sqrt{d}}, \qquad p_i = \frac{\exp(s_i)}{\sum_{j\in\mathcal{V}} \exp(s_j)}, \tag{1}$$

where d is the query–key dimension and $\mathcal{V}$ contains valid slots. Invalid padded keys are masked during attention and their output logits are excluded from normalization.

2.2 Structural properties

Permutation-equivariant scoring. In exact arithmetic, shared row-wise transformations commute with a candidate permutation matrix P. Self-attention obeys $\mathrm{Attn}(PH) = P\,\mathrm{Attn}(H)$ because query, key, and value rows, together with the key mask, are permuted consistently. Residual connections and row-wise normalization preserve this property. Consequently $s(PH) = Ps(H)$ and $p(PH) = Pp(H)$ when branch preprocessing and positions are unchanged and no candidate index is injected.

With a unique largest logit, permutation-equivariant scores preserve the selected candidate identity. For tied maxima, the selected candidate follows the first-slot tie-breaking rule. Floating-point reductions and changed batch shapes can also introduce numerical differences, so the mathematical property is distinct from bitwise reproducibility. Duplicate candidates are a different operation from permutation: adding another slot changes the softmax denominator and can change Set interactions.

Question isolation and candidate-set dependence. Separate questions occupy independent batch rows or separate inference calls. Read-only branch caches and question-local normalization and attention keep each question's scores independent of other questions. Within each question, Set attention conditions every candidate representation on the full candidate set. The relative probability $p_i/p_j = \exp(s_i - s_j)$ therefore depends on set-conditioned logits. Changing the candidate set can change the relative preference between two existing candidates while preserving isolation from other questions.

2.3 Training and probability calibration

Cross-entropy supervises the correct candidate on each example. We complement this supervision with a group-level confidence objective during training and scalar temperature fitting after training.

Calibration-aware training. We train the scorer with cross-entropy and a dataset-wise mean-confidence regularizer, adapting the confidence–accuracy alignment principle of DCA (Liang et al., 2020). For an effective batch partitioned into dataset groups $B(d)$, let $a_n = \arg\max_i p_{n,i}$, $c_n = \max_i p_{n,i}$, and $r_n = \mathrm{I}[a_n = y_n]$. Let $\mathcal{D}$ denote the datasets represented in the effective batch. We optimize

$$\begin{aligned} \mathcal{L}_{\text{train}} &= \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{conf}}, \\ \mathcal{L}_{\text{conf}} &= \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \left[\frac{1}{|B(d)|} \sum_{n \in B(d)} (c_n - \text{stopgrad}(r_n)) \right]^2 . \end{aligned} \tag{2}$$

Each group contributes equally. Balanced effective batches provide observations from every dataset, and the penalty aligns each group's mean confidence with its observed accuracy. Gradients flow through the selected probability; correctness supplies a detached target. The cross-entropy term retains supervision for individual candidate decisions. The training-loss ablation uses the same sampling and optimization settings with the penalty coefficient set to zero.

Reserved probability calibration. After fitting the scorer, we freeze its weights and fit a positive scalar temperature T on a separate calibration split by minimizing candidate-event NLL. Deployment uses $p_i^{(T)} = \exp(s_i/T)/\sum_{j \in \mathcal{V}} \exp(s_j/T)$. A shared scalar adjusts confidence while preserving score ordering, predicted candidate identity, and permutation equivariance. Temperature fitting uses the held-out task and candidate-count distribution.

Equivalent answer events. For the registered VQA protocol, blank and duplicate candidate strings are preserved. Prediction selects the maximum-logit slot, with first-slot tie breaking, before comparison by literal answer text. For probability scoring, the mass of identical strings is summed. Given a gold answer event G, the candidate-conditioned loss is

$$\mathcal{L} = -\log \sum_{i \in G} p_i = \text{LSE}_{j \in \mathcal{V}}(s_j) - \text{LSE}_{i \in G}(s_i). \tag{3}$$

Labels supply supervision to the loss and evaluator; forward prompts contain the image, question, and candidate text. This convention preserves the official answer-text task while keeping slot and answer-event probabilities distinct.

2.4 Shared execution

The branch structure repeats image and question prefixes across candidate encodings. We reuse these common computations while retaining each branch's original token visibility and question-local readout.

Shared-prefix inference. At inference time, the causal branch computation is factored into a shared prefix and independently evaluated suffixes. For multiple questions about the same image, the image-containing prefix is evaluated once and its key–value states are reused by the question and candidate branches. When requests also share a question, the reusable prefix extends through their identical question tokens. Prefix reuse requires matching token IDs, image features and multimodal positions. Each suffix receives a separate cache view and its original branch positions; request-local prefix states remain unchanged. Grouping compatible suffixes into a batch reduces repeated backbone calls. The resulting question and candidate representations are regrouped by question before applying Set attention and masked normalization. This execution schedule preserves the scorer's parameters and branch visibility. Prefix length, candidate count, batch shape, and cache memory determine execution cost. Numerical validation compares the resulting logits and decisions with ordinary branch execution under the same precision.

Tree-mask execution. An additional execution path packs the image prefix as a root, question prefixes as internal branches, and query/candidate suffixes as leaves. A token attends to its causal predecessors within its segment and to its ancestors. Question branches and sibling candidate leaves are isolated. Multimodal positions are gathered from the original unshared sequences, and leaf readouts feed the same question-local Set module and pointer. The current implementation uses dense masked attention, whose computation and memory depend on the packed sequence length. Tree packing and prefix-cache batching are alternative execution schedules for the same branch organization.

3 Experimental setup

All comparisons use Qwen3-VL-4B-Instruct (Bai et al., 2025). Vision parameters are frozen; language attention projections receive LoRA adaptation (Hu et al., 2021). We compare Set Pointer with a complete-candidate LM that reads all labeled alternatives and directly scores their answer tokens. Both directly output candidate probabilities. The main interface comparison combines candidate-conditioned cross-entropy and the dataset-wise calibration loss, optimized with AdamW (Loshchilov & Hutter, 2017). Independent Pointer is included in the relation test to probe the limitation of candidate-independent utilities. It uses independent candidate branches and a shared pointer readout. Section 4.1 details the input organization and scoring components of each interface.

Unified recognition. We train one checkpoint per method across Pets (Parkhi et al., 2012), CUB, Cars (Krause et al., 2013), Dogs, Flowers (Nilsback & Zisserman, 2008), and Aircraft (Maji et al., 2013). The cleaned training, development, and calibration inventories contain 31,088, 3,637, and 3,635 images. The standard test uses 35,566 FOCI-style four-choice records (Geigle et al., 2024) distributed with Finedefics (He et al., 2025), with identical shuffled candidates for all methods. We retain the author's repeated Dogs test row. Every question includes the gold class and a specified set of distractors, defining conditional multiple-choice classification. We reuse the published evaluation records and train both interfaces with our unified Qwen3-VL protocol.

Main model and component ablation. The main recognition and probability comparisons in Tables 3 and 6 use balanced hard-four-choice training. Each effective batch contains 96 questions, with 16 from each dataset. Both interfaces use seed 20261021 and the fixed epoch-three checkpoint after 2,016 updates, with identical candidate exposures. The full method uses calibration-loss weight five and a separate temperature minimizing equal-dataset calibration NLL. Table 7 isolates the training penalty and temperature within Set under this same recipe. Appendix A.1 specifies the optimization details.

Candidate-count study. The variable-count training ablation uses a separate unified CE recipe: three epochs with matched image exposure, alternating difficult and random distractors, and development selection by equal-dataset accuracy with earlier-epoch tie-breaking. Fixed-count training uses four candidates; variable-count training samples uniformly from $K \in \{2, 4, 10\}$. The reported count-transfer tests use difficult distractors at two, four, ten, and twenty candidates alongside the standard records. The four checkpoints share the training seed and test manifests. This study matches image exposure and updates; candidate-token computation varies with candidate count. Twenty-candidate testing measures transfer beyond the training count range.

Order and relation tests. The order test evaluates all 24 permutations of four candidates on 755 class-stratified questions, aligning candidate identities before comparison. Both this test and the difficult-candidate count-transfer evaluation use the complete-method checkpoints from Table 3, with unchanged weights and inference precision. Relation tests use synthetic scenes with paired candidate sets that reverse the required preference between shared objects. Independent, Set, and complete-candidate LM are evaluated with three training seeds at the same fixed final training endpoint on fresh scenes. The paired metric requires both questions correct; a largest-object control tests ordinary independent ranking.

Natural VQA. A separate supervised A-OKVQA experiment (Schwenk et al., 2022) evaluates ordinary task capability on an internal holdout of 2,576 questions, with disjoint training, selection, and calibration roles. We report the native LM comparison across three training seeds. Both interfaces are adapted separately to A-OKVQA. Split and optimization details are in Appendix A.

Probability quality and uncertainty. Each model receives its own scalar temperature fitted only on reserved calibration data (Guo et al., 2017). We report accuracy, NLL, sum-Brier, and ECE, distinguishing raw and scaled probabilities. Recognition ECE uses fifteen equal-count bins; the VQA protocol uses equal-width bins over literal answer events. Recognition intervals use 10,000 paired image-cluster bootstrap replicates within datasets and equal dataset weighting; repeated image rows stay together. They condition on the selected weights and fitted temperatures, excluding training-seed and calibration-fit uncertainty. Relation intervals resample whole scenes; the VQA interval additionally resamples the few available seeds and is exploratory. Intervals are marginal, without multiplicity correction. Further protocol details appear in Appendix A.

4 Results and analysis

4.1 Order stability

We first test whether the scoring interface preserves candidate identity under changes in presentation order. We compare three scoring interfaces, summarized in Table 1. Their inputs, candidate interactions, and readouts are defined as follows.

Independent Pointer. A shared Qwen3-VL encoder processes one image–question query branch and one image–question–candidate branch for each alternative. Each candidate branch receives only the image, question, and its own candidate text. Candidate summaries are projected to width 256 and scored by a shared query–key pointer, with softmax over valid candidates. Relative to Set Pointer, this baseline removes the cross-candidate Transformer blocks. Thus each candidate logit depends only on its own text, the image, and the question; other candidates affect its probability through normalization. Shared branch encoding and scoring yield permutation-equivariant scores.

Set Pointer. The proposed model uses the same branch organization and pointer readout, adding two position-free, pre-norm Transformer blocks over the 256-dimensional candidate representations. Each block uses four attention heads, a fourfold feed-forward expansion, GELU, and zero dropout. The question query bypasses the candidate-interaction stack. Self-attention allows candidate logits to depend on all alternatives, while shared row-wise operations and consistent masking preserve permutation equivariance.

Complete-candidate LM. The baseline encodes the image, question, and all labeled alternatives in a single sequence. Its pretrained LM head supplies the next-token logit for each single-token option label; these logits are normalized over the valid labels to directly obtain candidate probabilities. All candidates therefore share a sequential context, enabling joint comparison. Reordering options changes their positions and label assignments in this context, allowing scores to vary with presentation order. The permutation evaluation maps label predictions back to candidate identities before comparing decisions.

Table 1 contrasts the three architectures. The exhaustive permutation measurements below use the unified Set and LM checkpoints; Independent Pointer is additionally evaluated in the relation experiment in Section 4.2.

Interface	Candidate interaction	Equivariant scores
Independent Pointer	No	Yes
Set Pointer	Yes	Yes
Complete-candidate LM	Yes	Not guaranteed

Table 1: Structural properties under the preprocessing and masking conditions in the method. Equivariance is a score property; ties and floating-point effects require separate decision-level checks.

For the complete-method checkpoints used in Table 3, LM changes its selected candidate on 108 of 755 questions (14.30%) across exhaustive permutations; Set changes on none. Each question is evaluated under all 24 candidate orders, yielding 18,120 predictions per model. Predictions are aligned by candidate identity before comparison. Figure 2 connects this order stability to the complete-method performance under larger difficult candidate sets.

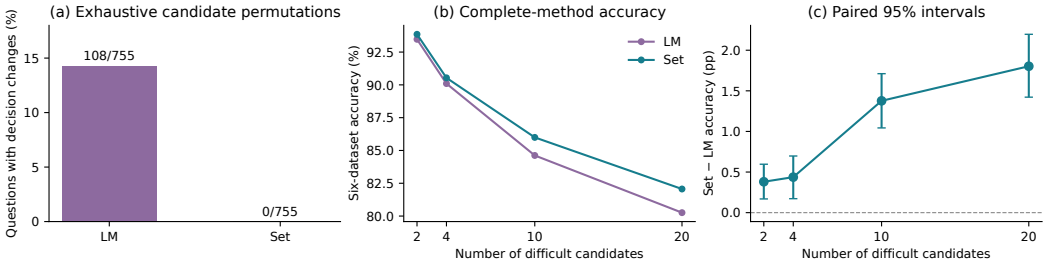


Figure 2: Complete-method structural analysis. (a) Fraction of questions whose predicted candidate identity changes across all twenty-four orders. (b) Six-dataset mean accuracy on difficult candidate sets. (c) Set-minus-LM accuracy with marginal paired image-cluster bootstrap intervals. All panels use the same complete-method checkpoints as Table 3; the training candidate count is four.

4.2 Candidate interaction enables preference reversals

We next test whether candidate interaction can express preferences that change with the candidate set. In each constructed pair, the middle-sized object among the candidates changes when an extreme candidate is replaced, reversing the required preference between two shared objects. A fixed independent utility assigns the same relative ranking to the shared objects in both questions. Set and complete-candidate LM can condition on the set. The fresh-scene test contains 512 scenes and 2,048 questions, with 1,024 questions and 512 pairs per task; it evaluates the fixed final checkpoints, averaging all three training seeds.

Set resolves the median relation. Independent achieves 99.7721% accuracy on the maximum control and 0.0000% paired accuracy on preference reversals. The Set-minus-Independent median-pair difference is 98.3073 percentage points, with paired whole-scene interval $[97.4609, 99.0234]$. The complete-candidate LM provides a second interaction-capable reference. The experiment demonstrates the role of candidate-dependent computation. The synthetic test reports raw probabilities.

Model	Median relation Acc. (%)	Pairs (%)	NLL	Maximum control Acc. (%)	NLL
Independent	50.0000	0.0000	0.70072	99.7721	0.01175
Set	99.0234	98.3073	0.25971	99.5768	0.03669
Complete LM	99.7721	99.5443	0.01878	99.8372	0.01062

Table 2: Fresh-scene test of fixed 80-epoch checkpoints, equally averaging the three retained training seeds. Pairs require both questions correct; all NLL values are uncalibrated. Set and complete-candidate LM both learn the median relation, which requires candidate interaction. The training budget is matched by optimizer updates.

4.3 Standard recognition and candidate-count transfer

We evaluate the complete decision model under matched balanced six-dataset training. Both interfaces use cross-entropy plus the dataset-wise calibration loss and a separately fitted temperature. Table 3 reports one mixed-dataset model per interface, using the fixed final checkpoint from one shared training seed. Set reaches 90.35% equal-dataset accuracy versus 90.09% for LM, a difference of +0.2690 percentage points with paired interval $[+0.0086, +0.5317]$.

Dataset	LM (%)	Set (%)	Difference (pp)
Pets	95.88	95.42	-0.46
CUB	84.52	85.57	+1.05
Cars	93.74	93.83	+0.09
Dogs	85.65	85.80	+0.15
Flowers	99.11	98.99	-0.11
Aircraft	81.61	82.51	+0.90
Equal-dataset mean	90.09	90.35	+0.27

Table 3: Unified mixed-dataset models on author-defined four-choice test records, using the same shuffled candidate sets for both methods. Each method uses one balanced CE-plus-calibration checkpoint across all six datasets, followed by its own held-out temperature. One training seed; differences are Set minus LM, computed before rounding.

Transfer to larger difficult candidate sets. We evaluate these same complete-method checkpoints at candidate counts of two, four, ten, and twenty, keeping the class inventory fixed. Distractors are selected by the recorded image–text ranking. Table 4 reports equal-dataset accuracy. The Set–LM gap is +0.44 percentage points at four candidates (paired 95% interval $[+0.17, +0.70]$) and +1.80 points at twenty candidates ($[+1.42, +2.20]$). The larger difficult sets therefore expose a stronger accuracy advantage for Set within this evaluation, while the model is trained with four candidates.

Candidates	LM (%)	Set (%)	Difference (pp)
2	93.47	93.85	+0.38
4	90.10	90.54	+0.44
10	84.62	86.00	+1.38
20	80.26	82.06	+1.80

Table 4: Difficult-candidate count transfer of the complete-method checkpoints in Table 3. Training uses four candidates; test candidates are mined with the fixed image–text ranking. Values are six-dataset mean accuracies from one shared training seed. Differences are Set minus LM.

4.4 Training with variable candidate counts

We test how exposure to variable candidate counts affects performance on larger decision sets. We train each interface with a candidate count sampled independently and uniformly from $K \in \{2, 4, 10\}$ for each image and epoch. Image exposure, optimizer updates, backbone and development-selection rule are held fixed; the standard four-choice development set still selects the checkpoint. Candidate-token computation varies with the sampled count. The test candidate sets are identical across all four checkpoints, and twenty candidates remain outside the training support. The category inventory is held fixed while candidate count varies. At twenty difficult candidates, the change from fixed to variable training is

Test	LM fixed	Set fixed	LM variable	Set variable
Standard	89.08	89.79	89.08	89.82
Hard 2	92.87	93.36	92.73	93.26
Hard 4	89.25	89.93	89.13	89.91
Hard 10	84.94	86.27	86.02	87.24
Hard 20	82.01	84.47	84.20	85.82

Table 5: Equal-dataset accuracy (%) for fixed four-candidate training and variable-candidate training. All four models use the same single training seed and test records. Difficult-distractor tests and the author-defined standard test are separate constructions.

+2.19 [+1.86, +2.53] percentage points for LM and +1.35 [+1.08, +1.62] for Set (paired image-bootstrap intervals). The corresponding change in the Set–LM gap is -0.85 [-1.25, -0.45] percentage points. These comparisons condition on the selected weights and assess the variable-count training recipe, including its candidate-token computation. Both methods improve at larger difficult candidate counts, but LM benefits more at twenty candidates, narrowing the Set–LM gap. Set retains a positive gap of +1.62 [+1.31, +1.93] percentage points after variable-count training. Thus the magnitude of the fixed-count advantage depends partly on training exposure. Standard-test point estimates remain close to their fixed-count counterparts. On the same exhaustive-order subset, the variable-count Set still has no decision changes, whereas the variable-count LM still changes candidate identity. The permutation test distinguishes the order behavior of the two interfaces after variable-count training.

4.5 Training and post-training calibration

Table 6 evaluates the probability outputs of the complete-method checkpoints. Independently fitted temperatures reduce NLL and Brier for both interfaces while preserving their decisions. After calibration, Set has NLL 0.2831 versus 0.2778 for LM; the paired Set-minus-LM difference is +0.00530 with interval $[+0.00004, +0.01054]$. Set has a lower Brier point estimate, 0.1428 versus 0.1438, with difference -0.00105 and interval $[-0.00356, +0.00148]$. Figure 3 visualizes how temperature scaling changes the confidence–accuracy relationship within each dataset.

	Raw		Temperature-scaled	
Metric ↓	LM	Set	LM	Set
NLL	0.3008	0.3478	0.2778	0.2831
Sum-Brier	0.1473	0.1480	0.1438	0.1428
ECE	0.0337	0.0448	0.0224	0.0355

Table 6: Equal-dataset mean probability metrics for the balanced CE-plus-calibration checkpoints in Table 3. ECE uses fifteen equal-count bins and is descriptive. Lower is better. Calibration uses post-training temperature scaling.

Training-time calibration. We further compare cross-entropy training with a dataset-wise mean-confidence penalty under a matched, balanced six-dataset training recipe. Both

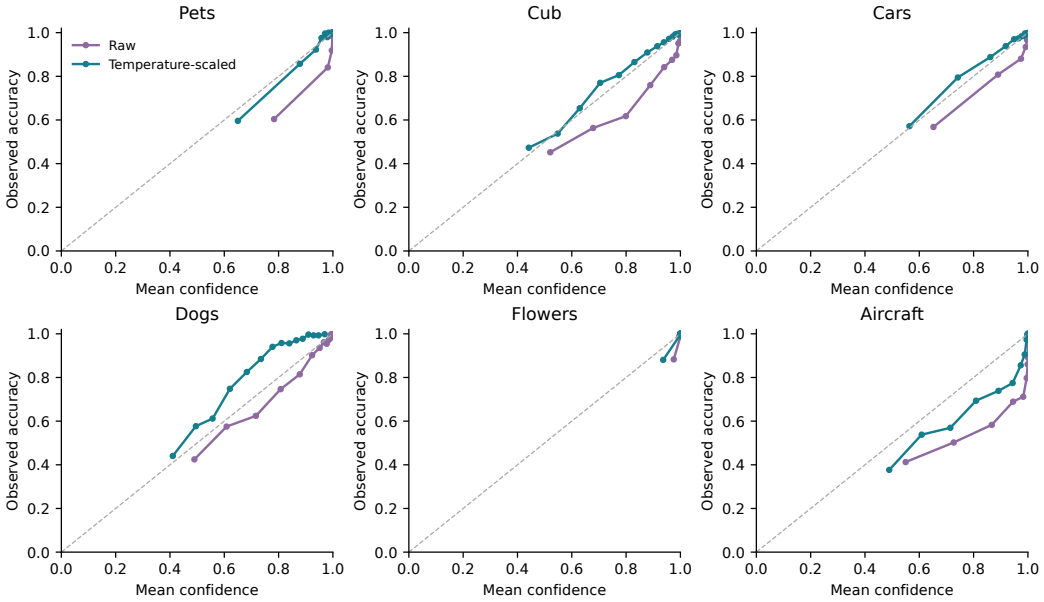


Figure 3: Probability calibration of the complete Set model on the six recognition datasets. Each curve uses fifteen equal-count confidence bins; points show mean top-choice confidence and observed accuracy. The dashed diagonal denotes matched confidence and accuracy. Temperature scaling uses the single scalar fitted on the reserved calibration data, with unchanged predictions.

Set models use the same seed, candidate exposures, and final-epoch checkpoint. Table 7 shows that the penalty improves raw NLL, Brier, and ECE while changing macro accuracy from 90.48% to 90.35%. Fitting a separate temperature for each model by minimizing equal-dataset calibration NLL further improves probability quality: NLL decreases from 0.2985 to 0.2831 and Brier from 0.1457 to 0.1428 when the training penalty is added. The paired NLL difference is -0.01539 with interval $[-0.01953, -0.01142]$; the accuracy difference is -0.1273 percentage points with interval $[-0.3205, +0.0676]$. These results support combining the training penalty with post-training temperature scaling. Appendix A.1 specifies the matched training recipe.

Training	Probability	Accuracy (%)	NLL ↓	Brier ↓	ECE ↓
CE	Raw	90.48	0.4003	0.1519	0.0521
CE + calibration	Raw	90.35	0.3478	0.1480	0.0448
CE	Temperature	90.48	0.2985	0.1457	0.0427
CE + calibration	Temperature	90.35	0.2831	0.1428	0.0355

Table 7: Training-loss comparison for Set under balanced six-dataset training with one shared seed. Metrics are equal-dataset means on the standard test; Brier is summed over candidates and ECE uses fifteen equal-count bins. Each model receives its own held-out temperature fitted with an equal-dataset NLL objective. Positive scalar temperature scaling preserves accuracy.

4.6 Natural VQA performance

Natural VQA tests performance outside fine-grained recognition. Table 8 compares the supervised native LM and Set on the same holdout. Set reaches 86.7495% accuracy versus 87.2800% for native LM: a mean decrease of 0.5305 percentage points. Thus the observed accuracy remains close to the LM baseline while using the Set decision interface. The

exploratory crossed seed/image interval for Set minus LM is $[-1.7041, 0.6406]$ percentage points; the interval provides a descriptive estimate of the accuracy difference under the registered protocol. Reserved calibration reduces average probability losses. The calibrated NLL and Brier point estimates remain higher for Set, complementing the accuracy comparison. Raw differences and statistical details appear in Appendix A.

Interface	Accuracy (%)	Scaled NLL	Scaled Brier
Native LM	87.2800	0.3703	0.1845
Set	86.7495	0.3819	0.1932

Table 8: Natural VQA capability and calibrated probability quality, using the completed supervised holdout experiment. Means use all registered seeds and frozen, independently fitted temperatures. The comparison follows the registered native-LM secondary analysis. Raw probability differences and protocol details appear in Appendix A.

5 Related work

Multimodal decision models. Recent Jev-style systems map runtime-defined alternatives to probabilities. Visual Jev and PixelJev adapt vision–language models using option-token readouts (Yu & Yao, 2026; Zhou et al., 2026), while Koa-action learns atomic label tokens with randomized label mappings (Dai et al., 2026). OmniJev combines learned option scoring with probabilistic supervision and calibration,[2] and Chinese-Jev develops a text-encoder decision model with heterogeneous supervision (Wang et al., 2026). Shared-context execution provides a complementary systems direction: Visual Jev reuses image prefixes, and FastJEV studies state reuse and candidate-layer pruning (Yu & Yao, 2026; Ma et al., 2026). Open-VisualJev focuses on the scoring architecture, combining independent candidate representations with explicit, permutation-equivariant joint comparison.

Set-dependent choice. Deep Sets establishes invariant and equivariant set representations, and Set Transformer models interactions through attention (Zaheer et al., 2017; Lee et al., 2019). Context-dependent choice models such as FETA and FATE allow an alternative's utility to depend on the surrounding set, including preference reversals when another alternative changes (Pfannschmidt et al., 2019). Our candidate module applies these principles to multimodal representations. The relation experiments test this capacity directly, alongside the stability of scores under candidate permutations.

Probability quality and decision evaluation. Jev benchmarks and decision-gate studies examine calibration, selective prediction, option ordering, and candidate-count robustness (Deußer et al., 2026; Rafe & Das, 2026). Judge routing, security screening, and mobile action selection illustrate the downstream use of decision scores (Li et al., 2026; Liu, 2026; Zhang, 2026). Visual MCQA debiasing and selective VQA likewise connect answer confidence with reliable selection (Atabuzzaman et al., 2025; Eisenschlos et al., 2024). Our evaluation separates recognition accuracy, order stability, and probability quality. The calibration pipeline builds on confidence–accuracy regularization and multiclass calibration objectives (Liang et al., 2020; Hebbalaguppe et al., 2022), followed by temperature scaling (Guo et al., 2017); ablations measure the contributions of training-time and post-training adjustment.

Fine-grained visual recognition. FOCI formulates fine-grained recognition as multiple-choice answering with difficult distractors (Geigle et al., 2024). Finedefics combines attribute-mediated alignment with classification instruction tuning (He et al., 2025). We reuse the candidate records released with the Finedefics evaluation code and train a unified Qwen3-VL model across six datasets. This provides a common recognition setting for comparing scoring architectures and complements the controlled structural experiments.

[2] OmniJev official repository, v1.1, accessed October 14, 2026.

6 Conclusion

Open-VisualJev separates shared candidate encoding, position-free set interaction, dynamic pointer scoring, calibration-aware training, and held-out temperature scaling into a multimodal decision pipeline. The design combines permutation-equivariant scores with preferences conditioned on the candidate set. Exhaustive permutations verify decision stability, and preference-reversal tasks demonstrate the expressive role of candidate interaction. Under the evaluated single-seed balanced protocol, Set achieves higher macro accuracy than the complete-candidate LM across six recognition datasets; dynamic candidate-count training improves both models on larger candidate sets. Separately supervised natural VQA yields accuracy close to the native LM. The balanced training-loss ablation supports combining mean-confidence regularization with temperature scaling to improve probability quality. These findings identify candidate-set structure as a useful design principle for multimodal decision models and provide an empirical basis for its use in dynamic choice tasks.

AI usage disclosure

We used generative AI tools throughout the research and writing workflow, under human direction. The writing workflow used the open-source `ai-research-writing-skill` to organize manuscript revision, evidence checks, and document validation. Assistance included literature search and synthesis; refinement of research questions, hypotheses, and the conceptual framework; formulation and exposition of mathematical properties; and design of experiments and statistical analyses. These tools also assisted with implementing model components, training and evaluation pipelines, data cleaning and reformatting, experiment orchestration, debugging, and result interpretation.

AI assistance contributed to the design and implementation of the procedural synthetic-scene generator used in the relation experiments. The resulting images, candidate sets, and labels were produced by executable generation code. Generative AI also assisted with creating the method illustration, drafting and restructuring the manuscript, English–Chinese translation, language editing, reference formatting, and document production.

The reported empirical results are computed from model predictions using the recorded evaluation and analysis scripts. Verification in the workflow includes checkpoint reload comparisons, data-manifest and artifact checks, numerical-evidence checks, citation checks, and inspection of rendered documents. The human author defines the research objectives, makes methodological and editorial decisions, and reviews the manuscript and supporting evidence. The author retains responsibility for the final text, claims, code, data, and figures, including AI-assisted contributions.

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A Essential reproducibility details

Architecture and scoring. The backbone hidden width is 2560. Candidate summaries are projected to width 256; Set uses two pre-normalized position-free Transformer blocks, four attention heads, feed-forward expansion four, GELU, and zero dropout. Independent removes cross-candidate blocks. The separate question query bypasses candidate attention. The readout operates in single precision over bfloat16 backbone features. The LM scores the next-token logits of single-token candidate symbols, directly normalized over the supplied alternatives. Pointer branches omit candidate indices. Pointer uses separate candidate contexts, while LM uses one complete-candidate context.

Recognition data and optimization. Pets, CUB, Cars, and Dogs reserve development/calibration images within classes from the official training-side inventory. Flowers and Aircraft use official validation images for those roles. Exact and reviewed near-duplicate groups are separated from test-side images; additional cross-dataset exclusions precede mixed training. The duplicate audit covers the task-specific training and evaluation inventories. The six datasets contain 37, 200, 196, 120, 102, and 100 classes, respectively; the choice count is independent of class count.

FOCI/Finedefics test candidate sets are preserved, including the original repeated Dogs row, and shuffled consistently across methods. The unified protocol follows the released Dogs test inventory. Training and dynamic tests mine difficult incorrect labels with the recorded OpenCLIP ViT-L-14 image–text ranking; random distractors are distinct sampled incorrect classes. The correct label is included. The standard author-defined four-choice test and the newly constructed difficult four-choice test are separate conditions.

Unified training uses learning rate 0.0001, accumulation over eight questions, three epochs, cosine decay, warmup fraction 0.03, AdamW weight decay 0.01, global clipping at norm one, and language-attention LoRA rank/scaling 16/32. The seed is 20261021. Both methods use candidate-conditioned cross-entropy, the same exposure and selection rule, and development-based checkpoint selection. Variable-count training changes only the registered candidate-count sampling and associated candidate construction; candidate-token computation follows the sampled candidate counts. Images use processor area bounds 65536 and 262144 pixels; Aircraft crops the bottom twenty pixels. Other recognition images are uncropped.

Relation construction and endpoint. Images contain colored objects with distinct sizes. Paired three-choice questions share two middle objects but replace the third candidate with a smaller or larger extreme, reversing the median answer. The largest-object task serves as an independent-ranking control. Training updates language LoRA and the trainable pointer head using choice-label supervision. The original training inventory has 256 scenes; the final test uses fresh scenes to evaluate the fixed training endpoint. Seeds are 20261011, 20261012, and 20261013. The common fixed endpoint is 80 epochs, using AdamW learning rate 0.0003, accumulation over 64 questions, weight decay 0.01, and clipping at norm one. This longer-budget test followed earlier development diagnostics; it evaluates the final achievable capability under this recipe. All final seeds are retained. Whole-scene bootstrap intervals condition on those trained seeds.

VQA splits and training. A-OKVQA training uses 12769 questions over 12404 images; selection uses 859 questions over 827 images, and calibration uses 852 questions over 827 images. The internal holdout has 2576 questions over 2482 images, grouped into 2481 resampling components after a reviewed near-duplicate merge. It is split from the official training inventory. Seeds are 20260924, 20260927, and 20260930. Models train for three epochs with learning rate 0.0002 and accumulation over eight questions; LoRA, optimizer, warmup, and clipping settings otherwise follow the shared recipe above. Checkpoint selection maximizes answer-text correctness, then minimizes NLL, then prefers the earlier epoch. Blank and duplicate choices are retained; probability mass for duplicate literal answer strings is summed when computing answer-event losses. Images are uncropped.

VQA comparison scope. The main text presents the registered native-LM secondary comparison. The original primary contrast used a canonically sorted LM input: Set-minus-canonical-LM accuracy was -0.5694 percentage points, with conditional interval $[-1.2861, 0.1289]$. The original primary and secondary analysis roles are retained. For native LM, raw Set-minus-LM NLL and Brier differences are $+0.1933$ and $+0.0243$; their exploratory crossed intervals are $[0.0779, 0.3260]$ and $[0.0078, 0.0407]$. Scaled differences are $+0.0116$ for NLL and $+0.0087$ for Brier, with crossed intervals $[-0.0054, 0.0284]$ and $[-0.0025, 0.0204]$. The registered analysis estimates accuracy and probability-loss differences with intervals.

Calibration and statistical conventions. Recognition minimizes reserved-set pooled candidate NLL over $\log T \in [-5, 5]$ using bounded scalar optimization and explicit boundary

checks. VQA uses a fixed grid of 2001 log-temperatures in the same range, preferring the value closest to zero on objective ties. Temperatures are fixed before test scoring. Sum-Brier sums squared probability errors across candidate or literal-answer events. Recognition ECE uses fifteen equal-count bins; VQA uses fifteen equal-width bins. Exact score ties are resolved by selecting the first slot. Paired bootstrap samples keep repeated images and related questions together. The recognition and relation intervals condition on the trained seeds; the crossed VQA analysis is exploratory with few training realizations. All intervals are conditional on the fitted temperatures and the evaluation distribution.

Reproduction scope. The recorded implementation uses PyTorch 2.8.0, Transformers 4.57.3, PEFT 0.17.1, and NVIDIA A800 devices. Data manifests, model checkpoints, raw predictions, and analysis scripts provide the evidence trail. The project archive preserves earlier specialist adapters, intermediate training endpoints, optimization probes, and engineering validation records. Public external reproduction remains a planned release milestone.

A.1 Training-time calibration protocol

The loss comparison uses balanced six-dataset hard four-choice training, matched candidate exposures, and a shared seed of 20261021. Each effective batch contains 96 questions, with 16 from each dataset. Both arms train for three epochs (2,016 updates and 193,536 question exposures) and use the final epoch-three checkpoint. The learning rate is 0.0001, with 61 warmup updates; the remaining optimizer and LoRA settings follow the recognition recipe. The coefficient of five is the evaluated development setting. The balanced recipe was developed after inspecting earlier benchmark results. Cross-entropy is augmented with a weight of five times the average dataset-wise squared gap between mean top-choice probability and empirical accuracy in the effective batch; the correctness target is detached. The control sets this weight to zero. Each model is evaluated on the same reserved calibration and standard test splits. Its positive scalar temperature minimizes equal-dataset mean calibration NLL. Paired image-cluster bootstrap intervals condition on these trained weights and fitted temperatures. Tables 3 and 6 compare the complete LM and Set interfaces under this same balanced recipe.